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Last login: Thu Jun 18 10:19:28 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd ~/Downloads/HPC-Upload\ 2/mpi-patterns
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_mirrored.c -o mpi_mirrored
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 2 ./mpi_mirrored 100000
Rank 0 (Outer-Left): Sending to mirrored Rank 1
Rank 1 (Outer-Right): Received from mirrored Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 2 ./mpi_mirrored 100000
Rank 0 (Outer-Left): Sending to mirrored Rank 1
Rank 1 (Outer-Right): Received from mirrored Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 4 ./mpi_mirrored 100000
Rank 1 (Outer-Left): Sending to mirrored Rank 2
Rank 0 (Outer-Left): Sending to mirrored Rank 3
Rank 2 (Outer-Right): Received from mirrored Rank 1 (First element: 10)
Rank 3 (Outer-Right): Received from mirrored Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_mirrored 100000
Rank 1 (Outer-Left): Sending to mirrored Rank 6
Rank 6 (Outer-Right): Received from mirrored Rank 1 (First element: 10)
Rank 3 (Outer-Left): Sending to mirrored Rank 4
Rank 2 (Outer-Left): Sending to mirrored Rank 5
Rank 0 (Outer-Left): Sending to mirrored Rank 7
Rank 5 (Outer-Right): Received from mirrored Rank 2 (First element: 20)
Rank 4 (Outer-Right): Received from mirrored Rank 3 (First element: 30)
Rank 7 (Outer-Right): Received from mirrored Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mkdir -p "/Users/stephenrajva/Downloads/SST_Stimulator/Communication_Pattern/Mirrored_experiment"

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cd "/Users/stephenrajva/Downloads/SST_Stimulator/Communication_Pattern/Mirrored_experiment"
stephenrajva@Stephens-MacBook-Air-4 Mirrored_experiment % head -100 ~/Downloads/HPC-Upload\ 2/mpi-patterns/mpi_mirrored.c
#include <stdio.h>
#include <stdlib.h>
#include <mpi.h>

```

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int main(int argc, char *argv[]) {
    int rank, size, N;
    int *data = NULL;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (argc != 2) {
        if (rank == 0) printf("Usage: mpirun -np <processes> %s <N>\n", argv[0]);
        MPI_Finalize();
        return 1;
    }

    N = atoi(argv[1]);
    if (size % 2 != 0) {
        if (rank == 0) printf("Error: Requires an EVEN number of processes.\n");
        MPI_Finalize();
        return 1;
    }

    data = (int *)malloc(N * sizeof(int));

    // --- PATTERN III LOGIC ---
    int half = size / 2;
    int partner = (size - 1) - rank; // Mirrored partner calculation

    if (rank < half) {
        // Left Side (Senders)
        for (int i = 0; i < N; i++) data[i] = rank * 10 + i;

        printf("Rank %d (Outer-Left): Sending to mirrored Rank %d\n", rank, partner);
        MPI_Send(data, N, MPI_INT, partner, 0, MPI_COMM_WORLD);
    } else {
        // Right Side (Receivers)
        MPI_Recv(data, N, MPI_INT, partner, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        printf("Rank %d (Outer-Right): Received from mirrored Rank %d (First element: %d)\n",
            rank, partner, data[0]);
    }
    // -----

    free(data);
    MPI_Finalize();
    return 0;
}

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stephenrajva@Stephens-MacBook-Air-4 Mirrored_experiment % cd ~/Downloads/HPC-Upload\ 2/mpi-patterns
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cp mpi_mirrored.c mpi_mirrored_original.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano mpi_mirrored.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_mirrored.c -o mpi_mirrored
mpi_mirrored.c:62:17: warning: format specifies type 'int' but the argument has
    type 'unsigned long' [-Wformat]

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```

59 |             "%d,%d,%d,%lf\n",
    |             ~~~
    |             %lu
60 |             rank,
61 |             partner,
62 |             N*sizeof(int),
    |             ~~~~~~
1 warning generated.
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano mpi_mirrored.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_mirrored.c -o mpi_mirrored
mpi_mirrored.c:62:17: warning: format specifies type 'int' but the argument has
type 'unsigned long' [-Wformat]
59 |             "%d,%d,%d,%lf\n",
    |             ~~~
    |             %lu
60 |             rank,
61 |             partner,
62 |             (unsigned long)(N*sizeof(int)),
    |             ~~~~~~
1 warning generated.
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano mpi_mirrored.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_mirrored.c -o mpi_mirrored
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _
Pattern/Mirrored_experiment/mirrored_trace.csv"
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 2 ./mpi_mirrored 100000
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pa
ttern/Mirrored_experiment/mirrored_trace.csv"
0,1,400000,0.000091
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _
Pattern/Mirrored_experiment/mirrored_trace.csv"
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 4 ./mpi_mirrored 100000
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pa
ttern/Mirrored_experiment/mirrored_trace.csv"
1,2,400000,0.000097
0,3,400000,0.000094
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _
Pattern/Mirrored_experiment/mirrored_trace.csv"
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_mirrored 100000
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pa
ttern/Mirrored_experiment/mirrored_trace.csv"
3,4,400000,0.000088
1,6,400000,0.000144
0,7,400000,0.000171
2,5,400000,0.000133
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns %

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